BM Byte Me

Project Risk Register

Risk Matrix and Usage

Likelihood \ Impact	Negligible	Minor	Moderate	Significant	Severe
Very Likely	Low Med	Medium	Med Hi	High	High
Likely	Low	Low Med	Medium	Med Hi	High
Possible	Low	Low Med	Medium	Med Hi	Med Hi
Unlikely	Low	Low Med	Low Med	Medium	Med Hi
Very Unlikely	Low	Low	Low Med	Medium	Medium

Figure 1: Risk Matrix [Boers([n. d.])]

The risk matrix shown in Figure 1 will be used to classify each risk that could arise during the development of the project. The categories range between 'Low', 'Medium', and 'High', with 'Low' requiring the least amount of mitigation, and 'High' requiring careful planning to make sure the risk does not become realised.

Risk Classification

Risks will be described by the following features: **1.** Description **2.** Likelihood **3.** Impact **4.** Severity **5.** Role **6.** Mitigation

Scope Creep

1. Features of the product becoming more complex without the additional time cost being considered. Leads to the work input to the project being much greater than expected.
2. Possible
3. Significant
4. Med Hi
5. Project Lead/Technical Lead
6. List of prioritised features that are highlighted each meeting. Clearly defined goals and a comprehensive success criteria.

Unconstructive Communication

1. Communication relating to the project that does not serve a distinct purpose. Can get stuck in loops of talking about the project instead of solutions within the project as well as including people that do not require as much depth in all communications.
2. Likely
3. Minor
4. Low Med
5. Project Lead
6. Create a focused plan for each meeting to ensure key points are discussed. Encourage communication on a need to know basis, clarifying the depth in which each role should be aware of the feature

Poor Resource Allocation

1. Inaccurately estimating the complexity of a task, which can lead to too many or too few personnel working at one time.
2. Unlikely
3. Moderate
4. Low Med
5. Project Lead
6. Assign appropriate story points to each task so the real complexity is understood. Dynamically monitor the progress of the task and adjust the allocation in stand-up meetings as needed.

Implementation Misunderstanding

1. Personnel assigned to a task not fully grasping the overall design of the feature, creating discrepancy in the interface being programmed to.
2. Unlikely
3. Moderate
4. Low Med
5. Technical Lead
6. Clear communication within the technical team regarding the design of the feature. Check ups when approving each pull request to ensure the design is followed.

Unrealistic Timelines

1. Inaccurately estimating the complexity of a task, which can lead to 'simple' tasks dragging on.
2. Possible
3. Significant
4. Med Hi
5. Documentation and Comms
6. Strong deadlines for functionality combined with the expected complexity that are dynamically adjusted in stand up meetings based on progress.

Data Security

1. Security inaccuracies that can allow private and sensitive data to be revealed without knowledge or permission.
2. Very Unlikely
3. Severe
4. Medium
5. Data Lead
6. Responsible storage of project API keys and strictly no sharing. Encryption/hashing on consumer data to ensure no leakage to a third party

Design Shortcomings

1. After Sprint 1 there is a chance that frameworks or design choices we have made will not scale well with our program as we expand its functionality. This may be because the original design does not integrate well with new the implementation of essential features.
2. Possible
3. Significant
4. Med Hi
5. Project Lead/Technical Lead
6. Identify vulnerable design choices at the start of Sprint 2. Programming to interface to avoid unnecessary rewriting of code. Future-proof these new design decisions to ensure maintenance is as simple as possible.

Bibliography

[Boers([n. d.])] Dane Boers. [n. d.]. ARMS Reliability: Beyond the risk matrix. https://www.armsreliability.com/page/resources/blog/beyond-the-risk-matrix